Manual

Expanded View of Quality Data / Alternative Inspection Start

AIP-EQD 8.2

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1 Expanded View of Quality Data / Alternative Inspection Start

Purpose

If you collect inspection data on the AIP, you can use the product *Expanded View of Quality Data / Alternative Inspection Start* to display additional quality data in parallel. Via configuration, you can display different quality data in the different input dialogs.

You can also activate the following special inspection modes:

- Goods Receipt
- Laboratory
- Calibration

Implementation notes

You use the product *Expanded View of Quality Data / Alternative Inspection Start* if you want to collect inspection data of any kind.

Integration

The product *Expanded View of Quality Data / Alternative Inspection Start* is part of the "data collection and information functions for quality data" and extends the inspection functions of the AIP shop floor terminal. Using this product, you can display additional quality data. You can also improve the inspection process and activate special inspection modes.

Features

The following functions are available:

- You can configure the display of additional quality data. This quality data is then displayed in parallel to the inspection data collection: List of documents, history of failures and measures, drawing, inspection note, control charts and histogram.
- You can define position and size of the additionally displayed quality data.
- Depending on the input type (variable, attributive, inspection chart, evaluation of codes, visual defects recording, etc.), you can display different objects.
- You can define globally valid configurations or configurations valid only for a terminal/terminal group.
- You can activate special inspection modes for the goods receipt, the laboratory tests and the calibration.

2 Expanded View of Quality Data

Purpose

Use the function "expanded view of quality data", if you want to show further quality data in parallel to the input dialog. This is for example necessary if you want to permanently display a stamped drawing in addition to the inspection data. You can show assigned documents as a list or optionally in full screen mode.

Requirements

Meet the following requirements:

- Active license AIP-EQD with AIP 8.2.
- ctaip.exe as of version: 8.2.0.24
- caq_dc_t.dll as of version: 8.2.0.13
- caq72.dll as of version: 8.2.0.3

The planned machine or machine group must be included in the group "Workplace" in the MOC application *Inspection requirement*, on the level of the inspection step, tab *Inspection step*. This requires that the option *One inspection step for each inspection station* is set in the inspection planning. On the level of the characteristics, no inspection stations need to be assigned.

Activation

To activate the option, make the following entry in the file "caq_dc_t.ini". This file is stored in the AIP subfolder "functions".

```
[OPTIONS]
ACTIVATE_EQD=1
```

Features

Use the expanded view of quality data to show additional context-sensitive information in the CAQ recording of measured values. Possible contexts are: measured value, inspection point or characteristic. The application provides the following display elements to show the information:

- Control chart
- Histogram
- Data list
  E.g. failures of a specific characteristic
- Document view

E.g. image of a part

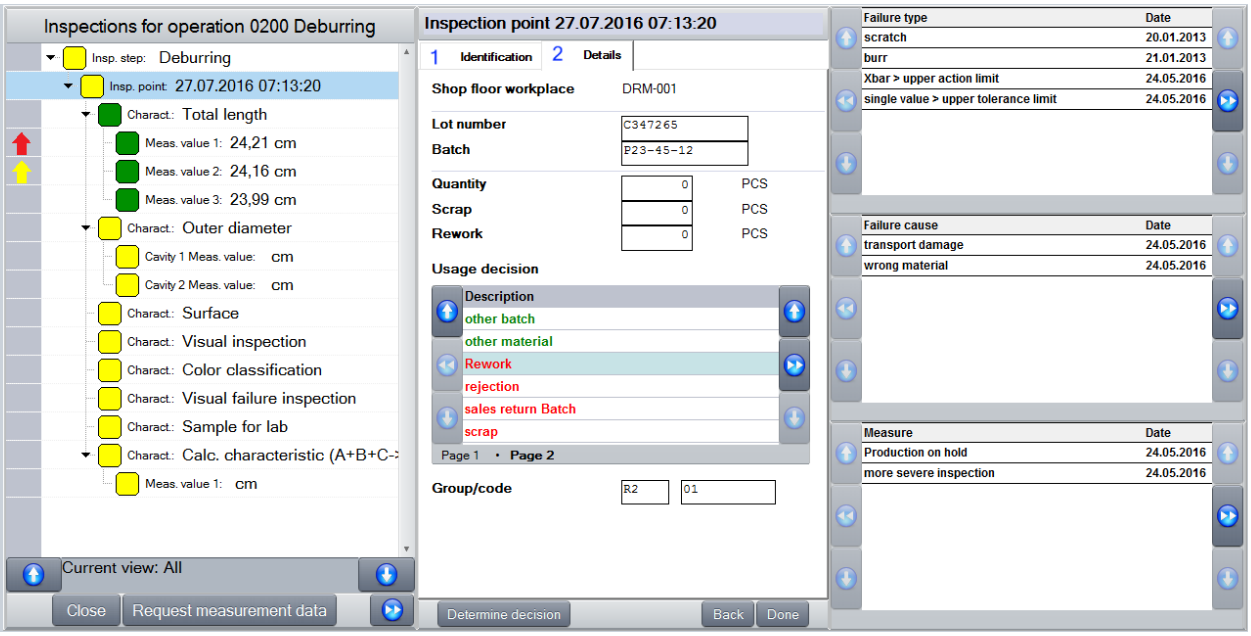
- Document list with document preview

  Configure a list of documents with direct document view.

- Document list without document preview

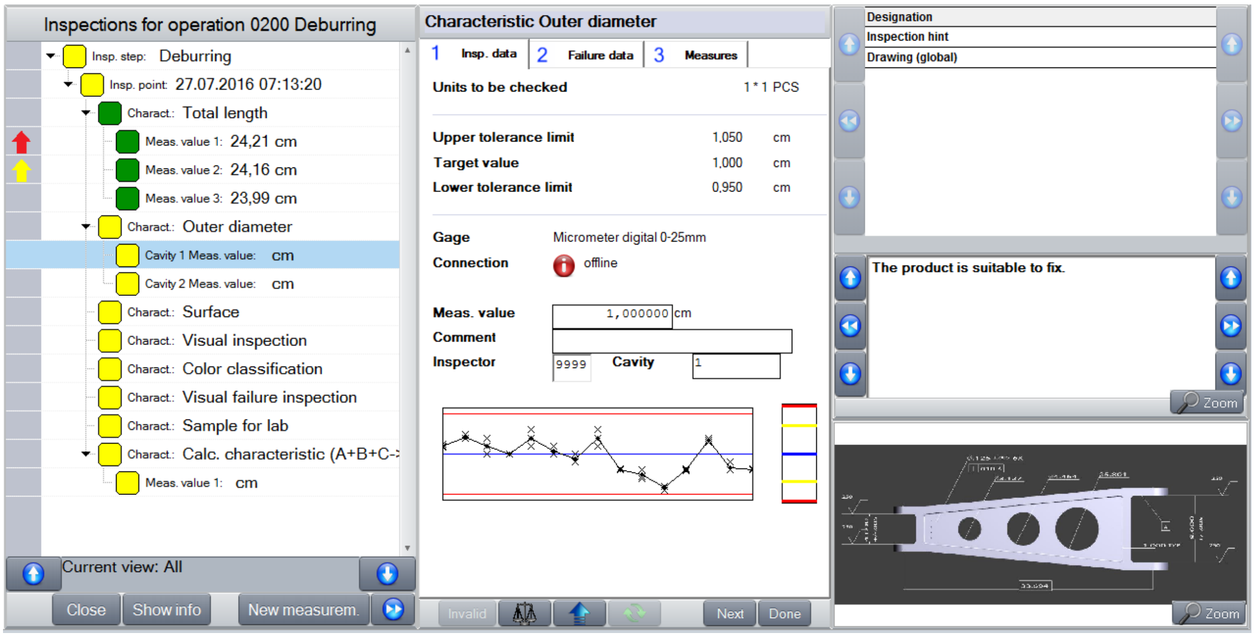
  Show a list and double-click a document that will then open in a separate window.

See the following layout examples:



Inspection points

<u>Showing measured values:</u>



Recording of measured values

Configure the display elements (e.g. control chart, histogram and data list) using a configurable number of rows and columns of the recording of measured values. Configure every display element separately for each input type. Example:

Show a list of failures for a characteristic and show a control chart for the measured values.

Use XML files for the configuration. Change specific properties of an element directly in the respective tag of the XML configuration.

Sample configuration for a display element

The below example shows a document list with direct display of the contents of the selected data record.

```
<element class="DocumentControl" col="5" colspan="2" row="1" rowspan="4" description="document list with displaying a document">
        <settings>
            <workdir>documents</workdir>
            <context_change>characteristic</context_change>
            <document_source>grid</document_source>
            <header_visible>false</header_visible>
            <view_align>vertical</view_align>
            <list_source>LIST;56</list_source>
            <list_dynamic_filter>RECTYP=%RECTYP%|BER=%BER%|PANNR=%PANNR%|PAUNR=%PAUNR%|AFO=%AFO%</list_dynamic_filter>
            <list_static_parameter></list_static_parameter>
            <columndef_section_name>QM-ducuments_EQD</columndef_section_name>
            <percentage_grid_usage>60</percentage_grid_usage>
            <can_zoom_document></can_zoom_document>
        </settings>
</element>
```

For further information refer to the document *Configuration_AIP-EQD.*

Integration of customer-specific components

In addition to the above-mentioned display elements and their configuration properties, you can also show customer-specific components in a specific display area of the AIP screen. To do so, the system sends the current node data to the component when you change nodes in the inspection list. Therefore, you can always refer to current CAQ data.

3 Alternative Inspection Start - Inspection without Operation Logon

Purpose

The following three fields of application have specific requirements regarding the inspection process. They are different to the usual in-production inspection on the production terminal.

- Incoming goods inspection
- In-production inspection at an exclusive inspection station (e.g. in the lab)
- Calibration

Use the Alternative Inspection Start for these three inspection processes. You do not need to log on an operation.

Requirements

Meet the following requirements:

- Active license AIP-EQD with AIP 8.2.
- ctaip.exe as of version: 8.2.0.33
- caq_dc_t.dll as of version: 8.2.0.15
- caq72.dll as of version: 8.2.0.7

Activation

The following fields of application have specific inspection modes:

- Incoming goods inspection
- In-production inspection at an exclusive inspection station (e.g. in the lab)
- Calibration

Only one of the modes can be active. Restart the terminal software if you change mode. For activation, one of the following configurations is required in the file „hytnrcfg.ini“ in the section „[CAQ->Optionen 0]“:

Inspection mode "Goods receipt inspection"

```
[CAQ->Options 0]
TERM_MODE= GOODS_RECEIPT
```

Inspection mode "Production inspection at an exclusive inspection station (e.g. in the lab)".

```
[CAQ->Options 0]
TERM_MODE=LAB
```

Inspection mode "Calibration"

[CAQ->Options 0]
TERM_MODE=CALIBRATION

The terminal uses the first entry found for "TERM_MODE".

If the parameter "TERM_MODE" in section "[CAQ->OPTIONS 0]" includes several entries, comment out the irrelevant ones.

Activate a specific main layout for the tile view of each of the inspection modes. The layout files can be found in the AIP subfolder "gui". Activate the main layout as follows:

Inspection mode "Goods receipt inspection"

Rename the file "l_main#lgoods#.xml“ in "l_main.xml".

Inspection mode "Production inspection at an exclusive inspection station (e.g. in the lab)".

Rename the file "l_main#llab#.xml“ in "l_main.xml".

Inspection mode "Calibration"

Rename the file "l_main#lcal#.xml“ in "l_main.xml".

In addition, the CAQ option *1157* must be enabled and have the entry *Y* in the field *Value.* Also assign the parameter *[GROUP:GEPLANT,MNR,MGRP,OPT_PLAN]* in the field *Addition*. The parameter *[GROUP:GEPLANT,MNR,MGRP,OPT_PLAN]* groups the inspection step characteristics by assigning them to the fields "Planned", Machine group", "Machine" and "Inspection station". For each combination an inspection step is generated. The inspection steps will then include the contents of the grouping criteria in the respective fields. The inspection steps must include these grouping criteria. The system uses the grouping criteria to specify the respective inspection steps that are required for the three special inspection modes.

The system creates an inspection step, if option *1157* is configured accordingly for each combination of grouping criteria and if the option *One inspection step for each inspection station* is enabled in the inspection plan header.

For the inspection mode *Calibration,* you must set the parameter PAN_AU/AUNR_COPY in the field *Action* of the application *Area-Order type configuration* (Menu entry *System administration* → *System settings*). This configuration automatically generates a calibration inspection requirement if the calibration calendar includes a calibration order.

Logging of data

If logging is required, you must set the following parameter in the hytnrcfg.ini in section "[OPTIONS 0]".

TERM_MODE_DEBUG=ON

Using this parameter, the inspection data included in the spool directory is not deleted when the inspection dialog is closed. This is required in case of an upload with this inspection mode, for example.

It is best to set this parameter in the local file hytnrcfg.ini, to write-protect the file and to restart the AIP. After the secured logging (e.g. via upload), remove the write-protection, delete the parameter and restart the terminal.

Functionality – Inspection mode "Goods receipt inspection"

The inspection mode "Goods receipt inspection" is displayed in a special layout.

All possible inspection steps of this workplace are displayed. The displayed inspection steps are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the machine group matches the one of the inspection station. In addition, the inspection steps planned for the machine are identified, if the machine and the inspection station are identical.

As soon as the system generates a goods receipt inspection via interface in HYDRA, the inspection is displayed after a cyclic update.

You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

```
[DATACONTEXT_GOODS_RECEIPT]
LOADCYCLE=300
DATAPROVIDER_ID=PAUMNR
LIST=u_l_caq_inspstep_tnr
```

The standard layout can display up to 18 inspection steps at the same time. You can integrate the filter field and then filter by all displayed inspection step contents. By default, the displayed inspection steps are sorted in ascending order according to the date/time of inspection step generation. You can use the "caliper" button to access the inspection process.

The function to complete the inspection requirement is a special feature. This function automatically identifies inspection steps and inspection requirement results. If required, you can change them.

When you complete an inspection step, the system automatically sets the QM operation to "completed". Requirement: the QM operation must not be logged on and the processing code must define the OP as *inspection OP*. If the finished QM operation is the last operation to be finished in the order, the order is then set to the status "Completed".

If the inspection step is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on when the terminal is online again ("posting required"). If the required posting is not made, the inspection step is not completed. There is no message that the inspection step has not been completed.

Functionality – Inspection mode "In-production inspection at an exclusive inspection station"

This inspection mode does not require an operation logon either. Contrary to the in-production inspection at an exclusive inspection station requiring a QM logon, this inspection mode instantly shows all possible inspection points at this inspection station.

All possible inspection points of this workplace are displayed. The displayed inspection points are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the machine group matches the one of the inspection station. In addition, the inspection steps planned for the machine are identified, if the machine and the inspection station are identical.
4. For each of the identified inspection steps, all possible inspection points are identified.

As soon as an inspection point is generated when an inspection is due, the terminal shows the inspection point of the specific inspection station after the specified updating interval.

You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

[DATACONTEXT_LAB]
LOADCYCLE=300
DATAPROVIDER_ID=PPKTMNR
LIST=u_l_caq_insppoint_tnr
SECTION=DATACONTEXT_LAB_INSP_PT_MATURITY

In the list of the inspection point tiles, you can manually generate free inspection points using the "+" button. Using the button, the relevant inspection steps of this inspection station are identified and displayed in a list you can filter. Select an inspection point in the list and use the button "Generate inspection point". A free inspection point is generated for this inspection step. If you do not want to generate further inspection points, close the list of inspection steps manually.

Inspection steps can be completed in this inspection mode. Focusing on detail information on Workplace/Machine, you can complete an inspection step using the button "Complete". The relevant inspection steps of this inspection station are identified and displayed in a list you can filter (see above: generating an inspection point). Select an inspection step and use the button "Complete inspection step". When you complete an inspection step, the system automatically sets the QM operation to "completed". Requirement: the QM operation must not be logged on and the processing code must define the OP as *inspection OP*. If the finished QM operation was the last operation to be finished in the order, the order is then set to the status "Completed". If you do not want to generate further inspection points, close the list of inspection steps manually.

If the inspection step is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on while the terminal is online again ("posting required"). If the required posting is not made, the inspection step is not completed. There is no message that the inspection step has not been completed.

Functionality – Inspection mode "Calibration"

This inspection mode automatically produces a calibration inspection requirement including respective inspection steps when a calibration order is generated in the calibration calendar. If you want to generate a calibration inspection requirement via calibration order, the following conditions must be fulfilled:

- Configure the calibration inspection plan as follows:
    - Operation assignment: “One inspection plan for all operations"
    - IO + inspection station: "One inspection step (=IO) for each inspection station"
    - Generate new QM OPs: "None"
    - IO + generate characteristics: "When generating inspection requirements"

- The inspection plan characteristics for the calibration must be planned for a machine/machine group.

The inspection mode "Calibration" has a specific layout. The displayed inspection steps are identified as follows:

1. The machine group of the inspection station is identified.
2. Using the specifications in the terminal configuration, the possible inspection steps are limited to the area type and the area.
3. The inspection steps of the machine group are identified, if the the machine group matches the one of the inspection station. In addition, the inspection steps planned for a machine are identified if the machine is the one of the current workplace.
4. You can configure the update interval in the file "caq72.ini" in the subfolder "packets" using the parameter "LOADCYCLE".

These inspection steps are instantly displayed in the inspection mode "Calibration" without operation logon.

```
[DATACONTEXT_CAL]
LOADCYCLE=300
DATAPROVIDER_ID=PAUMNR
LIST=u_l_caq_inspstep_tnr
```

A special feature of the inspection mode "Calibration" is that you can complete the calibration directly on the AIP when finishing the inspection step and requirement. In the process, you also select the test equipment status, the calibration result and the basis for calculating the next calibration.

If you complete the inspection requirement, the calibration order is automatically completed, too. As a condition, the respective QM operation must not be logged on and the OP must be defined as *inspection OP* via processing code.

The system resets the calibration in the calibration calendar in the same way as a maintenance activity.

If the inspection requirement is completed in the offline status, the operation is not automatically set to "completed". This would require a manual posting later on while the terminal is online again ("posting required"). If the required posting is not made, the inspection requirement is not completed. There is no message that the inspection requirement has not been completed.

4 Configuration for the Expanded View of Quality Data

	• →	

Purpose

Use this function, if you want to show further data in addition to the input dialog in the AIP inspection data collection. For example, you can show the list of documents or directly show a drawing.

The expanded view of quality data is based on a customizable layout for the inspection data collection.

Requirements

The functions described below are available for HYDRA 8 with AIP 8.2 service pack 10 and higher. You also need the license AIP-EQD.

For final activation, set the following parameter in the CAQ configuration file *caq_dc_t.ini* (folder: *functions)* in the *[OPTIONS]* section:

ACTIVATE_EQD=1

Basic configuration

Without the expanded view of quality data, the AIP inspection data collection always shows both: the inspection list and the dialog pertaining to the entry selected in the inspection list. The inspection list is always on the left-hand side. The relevant input dialog is always to the right of it. Use the expanded view of quality data to arrange these two objects individually. In addition to this, you can also show further quality data simultaneously. You can also arrange these additional objects according to your requirements.

Define the corresponding basic configuration in the file *caq_global_gui_settings.xml*. This file is stored in the AIP sub-folder .*functions*.

Definition of grids

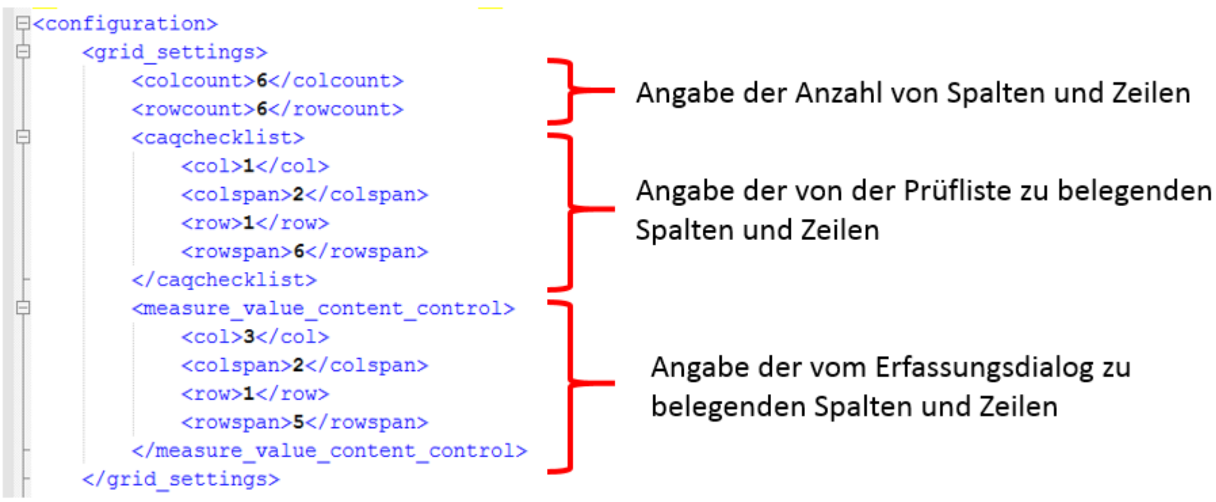
In the *<grid_settings>* section of this file, first divide the screen area into the required number of columns and rows.

Layout of inspection lists and dialogs

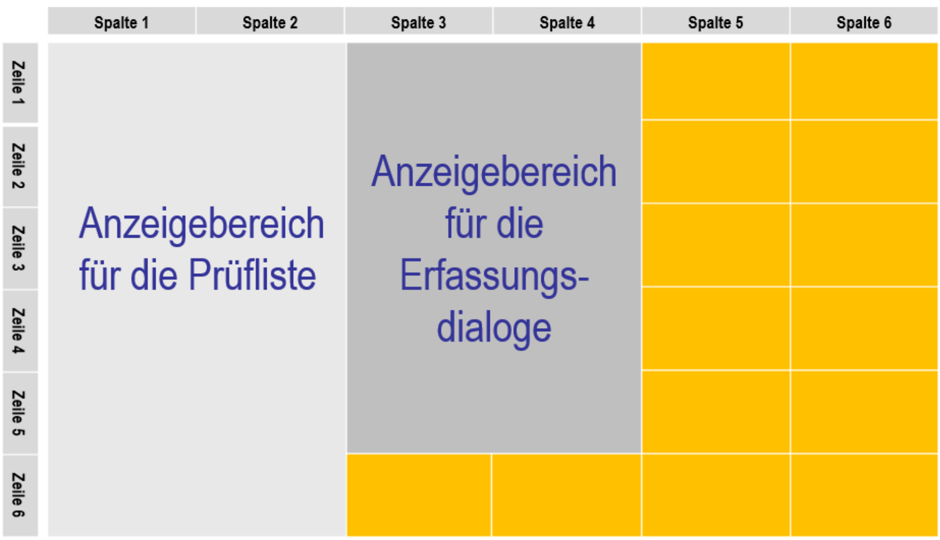
After the grids have been defined, specify the required columns and rows in the section <*caqchecklist*> to define which cells are populated by the inspection list. Define the display area for the input dialog in the <*measure_value_content_control*> section.

Use the parameters *col* (columns) and *row* (rows) to define where the cells that are populated begin. The *colspan* and *rowspan* parameters define the number of columns and rows that are populated. If the column and row width equals 1, you do not have to specify the *span* parameter.

The following illustration shows a configuration example.



The example configuration above would result in the following layout for inspection list and input dialogs.



Context elements

You can use the section <*context_elements*> to define context elements with a different database. For example, you require the definition of context elements to specify which change of a node or element does not trigger an update of the contents of the objects displayed additionally in the inspection list. For example, the document list or other objects need usually not be updated when you change from measured values container 1 to the next measured values container within the same characteristic. If you suppress an update, you can enter further data within a shorter time. Therefore, we recommend this data suppression.

To define a context element, make the following entry in the <*context_elements*> section.

<context_element id="context name">

The context name can be any name. Below the context element, define the parameters that clearly describe this context. With regard to the example about updating above, this means that the system triggers an update anytime the content of these parameters changes. Define the context parameters as follows:

<fields>Parameter1|Parameter2|Parameter3</fields>

The following parameters are available, among others.

- RECTYP e.g. WEP, FEP, WAP
- BER e.g. Areas of the inspection requirement
- PANNR Inspection requirement number
- PAUNR Inspection step number
- AFO Operation sequence number
- WERTNR Measured value number
- STPRNR Sample number
- EINTNR Inspection point number

Use a "|" to separate multiple parameters.

Below is an example of a configuration for the contexts *NO_RELOAD**, *characteristic*, *measured_value*, *inspection_point* and *sample*.

```
<context_elements>
    <context_element id="static">
        <fields>NO_RELOAD</fields>
    </context_element>
    <context_element id="inspection_step">
        <fields>RECTYP|BER|PANNR|PAUNR</fields>
    </context_element>
    <context_element id="characteristic">
        <fields>RECTYP|BER|PANNR|PAUNR|AFO</fields>
    </context_element>
    <context_element id="measure_value">
        <fields>RECTYP|BER|PANNR|PAUNR|AFO|WERTNR</fields>
    </context_element>
    <context_element id="inspection_point">
        <fields>RECTYP|BER|PANNR|PAUNR|EINTNR</fields>
    </context_element>
    <context_element id="sample">
        <fields>RECTYP|BER|PANNR|PAUNR|STPRNR</fields>
    </context_element>
</context_elements>
```

The *static* = *NO_RELOAD* element is an exception. This element can be used to completely disable the update of components.

Placeholders

Using the *context_placeholder* element, you can define separate placeholders, which are then replaced with the relevant values from the CAQ node data when setting up the list parameters.

For example:
Defining placeholders:

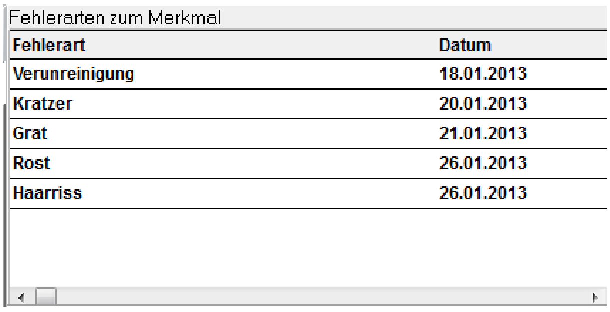
```xml
<context_placeholders>
    <placeholder>
        <name>%RECTYP%</name>
        <value>RECTYP</value>
    </placeholder>
    <placeholder>
        <name>%AFO%</name>
        <value>AFO</value>
    </placeholder>
    <placeholder>
        <name>%PANNR%</name>
        <value>PANNR</value>
    </placeholder>
    <placeholder>
        <name>%PAUNR%</name>
        <value>PAUNR</value>
    </placeholder>
    <placeholder>
        <name>%BER%</name>
        <value>BER</value>
    </placeholder>
    <placeholder>
        <name>%ATK%</name>
        <value>ATK</value>
    </placeholder>
</context_placeholders>
```

In a specified configuration, these placeholders replace the placeholder *%RECTYP%* with the RECTYP value from the current CAQ node data, for example.

Options

Define global settings in the <*options*> section. These settings serve as the default values in the following dialog configuration. If you defined settings in the "options" section, you do not have to repeat these configurations in the dialog configurations.

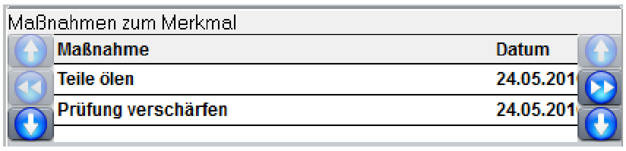
Currently, you can globally define the scroll mode for lists and the zoom option for documents. The available scroll modes are *scrollbars* and *buttons*.



Fehlerarten zum Merkmal

Fehlerart	Datum
Verunreinigung	18.01.2013
Kratzer	20.01.2013
Grat	21.01.2013
Rost	26.01.2013
Haarriss	26.01.2013

Mode: *scrollbars*



Maßnahmen zum Merkmal

Maßnahme	Datum
Teile ölen	24.05.201
Prüfung verschärfen	24.05.201

Mode: *buttons*

Use the *true/false* parameter to enable/disable the zoom function. If the zoom mode is activated, the 

icon appears on the bottom right in the document/document text displayed.

To scale the measured value input dialog, use the *default_scale_factor* option to change the calculated scaling factor subsequently. 100 equals the calculated factor.

You can change the design of the inspection list to a certain extent. If you want to change the design of the inspection list, use the parameter "*[FONT_VIRTUALSTRINGTREE] → DefaultNodeHeight*" of the configuration file *caq_dc_t.ini*.

For further information, refer to the relevant configuration description.

To request control chart and histogram data, you can use different data bases. They are configured in the following options:
- *default_controlchart_context*
- *default_histogram_context*

If you want to request a control chart for one characteristic, define the following fields:
RECTYP|BER|PANNR|PAUNR|AFO

If the control chart should be requested for an article for all orders, define the following fields:
RECTYP|BER|ATK|AFO

You can also define the data basis for each control chart or histogram in the *chart_data_source* configuration.

The following illustration shows a possible configuration:

```
<options>
    <option id="default_scale_factor">85</option>
    <option id="global_scroll_mode">buttons</option>
    <option id="can_zoom_document">true</option>
    <option id="default_controlchart_context">RECTYP|BER|PANNR|PAUNR|AFO</option>
    <option id="default_histogram_context">RECTYP|BER|PANNR|PAUNR|AFO</option>
</options>
```

Dialogs

You assign layout configurations to the relevant input dialogs in the *<dialogs>* section.

Specify the dialog as follows:

<dialog id="dialog specification">

The dialog used to collect measured values including a reference to the inspection point is, for example: *qee_mw_me_es_pp_si* and requires the following configuration. Use lower case letters for the specification.

<dialog id="qee_mw_me_es_pp_si">

The dialog specifications are included in the MOC application of the dialog configuration. The relevant input dialogs for inspection data generally start with *QEE*.

The above-mentioned configurations are always created for the first dynamic dialog of the relevant workflow configuration.
In the example above, the relevant input workflow is *MW_ME_ES_PP_SI*. Its first dynamic dialog has the name *QEE_MW_ME_ES_PP_SI*.

After the dialog has been specified, assign the layout XML file as follows:

<configuration_file>layout.xml</configuration_file>

You can define any XML file name.

```xml
<dialogs>
    <dialog id="qee_inspstep">
        <configuration_file>inspection_step.xml</configuration_file>
    </dialog>
    <dialog id="qee_insppoint">
        <configuration_file>insppoint.xml</configuration_file>
    </dialog>
    <dialog id="qee_sample">
        <configuration_file>insppoint.xml</configuration_file>
    </dialog>
    <dialog id="qee_mw_me_es_pp_si">
        <configuration_file>variable.xml</configuration_file>
    </dialog>
    <dialog id="qee_mm_me_es_pp_si">
        <configuration_file>characteristic.xml</configuration_file>
    </dialog>
    <dialog id="qee_mw_me_es_pp_ca">
        <configuration_file>variable.xml</configuration_file>
    </dialog>
    <dialog id="qee_mm_me_es_pp_ca">
        <configuration_file>characteristic.xml</configuration_file>
    </dialog>
    <dialog id="qee_mm_co_st_pp_si">
        <configuration_file>attributive.xml</configuration_file>
    </dialog>
    <dialog id="qee_mm_be_st_pp_fs">
        <configuration_file>inspection_chart.xml</configuration_file>
    </dialog>
</dialogs>
```

General element configurations

Configuration	Description	Valid values
context_change	Collection of acronyms causing the element to be refreshed.	The applicable values result from the definition of all context change methods of the global configuration.
workdir	Use this parameter to define a working directory in the corresponding CAQ order folder.	Use a valid folder name. If this value is not defined, the system assigns a name. The "workdir" specifications must be unambiguous within an XML file. This means that each element must have a different "workdir" specification. Static elements must have a "workdir" specification that is unambiguous with all XML files.

To display an element, you must or you can set the following attributes in the configuration for the element:

Configuration	Description

class	Defines the element class. The system currently supports the following element classes: - *ControlChart1* - *ControlChart2* - *Histogram* - *DocumentControl* - *Grid* - Custom class
col	Defines the column showing the element.
row	Defines the row showing the element.
colspan	(Optional) Defines the number of columns that should be displayed.
rowspan	(Optional) Defines the number of rows that should be displayed.
description	(Optional) Use this attribute to give the element a name. Then this name is displayed in the validation program if an error occurs.

Sample configuration:

```
<element class="Grid" col="5" colspan="2" row="1" rowspan="2" description="Failure types over all characteristics for the inspection step">
    <settings>
        <workdir>FailureTyps_InspectionStep</workdir>
        <context_change>inspection_step</context_change>
        <header_title></header_title>
        <header_visible>false</header_visible>
        <list_source>CPAUERR.LIST</list_source>
        <list_dynamic_filter>CPAUERR.RECTYP=%RECTYP%|CPAUERR.BER=%BER%|CPAUERR.PANNR=%PANNR%|CPAUERR.PAUNR=%PAUNR%</list_dynamic_filter>
        <list_static_parameter>CPAUERR.ERRTYP=FA</list_static_parameter>
        <columndef_section_name>QM-failure_type_EQD</columndef_section_name>
    </settings>
</element>
```

Element configurations to request data from the HYDRA server

Elements, which request data from the HYDRA server, require additional configurations. Include the following parameters for this purpose:

Configuration	Description	Valid values

list_source	Configuration of the list requested	See HYDRA lists
list_dynamic_filter	Dynamic filters to request data from the HYDRA server. The system then fills the specified acronyms with the values of the current node. e.g.: *RECTYP=%RECTYP%* *becomes: RECTYP=FEP*	All elements of the global configuration for the element *<context_placeholder>.*
list_static_parameter	Static parameters that are directly attached to the command for the HYDRA server. If you want to enable a combined filtering of a field content, e.g. showing failure types and failure causes in one grid, then separate them using a pipe slash ("|").	

Configuration example:

```
<element class="Grid" col="5" colspan="2" row="1" rowspan="2" description="Failure types and causes over all characteristics for the inspection point">
    <settings>
        <workdir>FailureTyps_InspectionPoint</workdir>
        <context_change>sample</context_change>
        <header_title></header_title>
        <header_visible>false</header_visible>
        <list_source>CPAUERR.LIST</list_source>
        <list_dynamic_filter>CPAUERR.RECTYP=%RECTYP%|CPAUERR.BER=%BER%|CPAUERR.PANNR=%PANNR%|CPAUERR.PAUNR=%PAUNR%|CPAUERR.STPRNR=%STPRNR%</list_dynamic_filter>
        <list_static_parameter>CPAUERR.ERRTYP=FA|FU</list_static_parameter>
        <columnheader_section_name>QM-failure_EQD</columnheader_section_name>
    </settings>
</element>
```

Control chart component

The class names are *ControlChart1* and *ControlChart2*. The numbers 1 and 2 refer to the control charts 1 and 2 you can define in the MOC.

Configuration	**Description**	**Default value***	**Valid values**
header_font_style	Header is shown in bold and/or italic		italic bold italic, bold
show_y_axis_desc	Shows the Y-axis	false	true false
show_x_axis_desc	Shows the X-axis	false	true false
show_specified_value	Shows the target value	false	true false

show_tolerance_limits	Shows tolerance limits	true	true false
header_visible	Shows header	false	true false
show_chart_caption	Shows the control chart description	true	true false
scale_by_y_axis	Scaling based on the Y-axis	false	true false
show_single_values	Shows each of the measured values	false	true false
show_grid_lines	Shows the grid Restriction: The grid is only shown if you enabled the display of both axis labels.	false	true false
workdir	Temporary working directory		
show_action_limits	Shows action limits	true	true false
header_title	Caption		
show_warning_limits	Shows warning limits	false	true false
measure_value_count	Number of measured values to be displayed	10	
background_color	Background color	R=255,G=255,B=255	

x_axis_field	X-axis label	Sample	All acronyms of file "RGK.lst" Special acronyms see (1) below the table
chart_data_source	Use these fields to request the relevant control chart data from the HYDRA server. If no definition is specified here, the system uses the values of the global option <default_controlchart_context>.	RECTYP|BER|PANNR|PAUNR|AFO	
type	Use this parameter if, as opposed to the control chart definition, you want to show another control chart in inspection planning. The feasible values are the active status IDs of the status type *REGELKART.* Refer to the status application to view these status IDs.		

*The default value will be used if you did not explicitly define a configuration. Therefore, you do not have to configure these properties with the default value.

(1) For the following acronyms, specially formatted outputs are available.

Acronym from RGK.lstt	Formatted acronym	Output format
WERT:DAT	DatumE	dd.mm.yyyy
WERT:ZEI	ZeitE	hh:mm
LWERT:DAT	DatumL	dd.mm.yyyy
LWERT:ZE	ZeitL	hh:mm
	Datum Zeit E Combined using DatumE + ZeitE	
WERT:VONK	Datum Zeit L Combined using DatumL + ZeitL	
SPABS:ZEI	ZeitA	hh:mm
SPABS:DAT	DatumA	dd.mm.yyyy
	Datum Zeit A Combined value from DatumL + ZeitL	

Sample configuration:

```
<element class="ControlChart1" col="3" colspan="1" row="6" description="Control chart 1">
    <settings>
        <workdir>ControlChart_1</workdir>
        <context_change>characteristic</context_change>
        <show_single_values>true</show_single_values>
        <show_specified_value>true</show_specified_value>
        <show_x_axis_desc>false</show_x_axis_desc>
        <show_y_axis_desc>true</show_y_axis_desc>
        <show_tolerance_limits>true</show_tolerance_limits>
        <show_action_limits>true</show_action_limits>
        <show_warning_limits>true</show_warning_limits>
        <show_chart_caption>true</show_chart_caption>
        <show_grid_lines>false</show_grid_lines>
        <background_color>R=213,G=224,B=245</background_color> <!--for example light blue: R=213,G=224,B=245-->
        <header_title>R</header_title>
        <header_visible>false</header_visible>
        <header_font_style>bold,italic</header_font_style>
        <measure_value_count>20</measure_value_count>
        <chart_data_source></chart_data_source>
        <type>XMED</type>
        <x_axis_field></x_axis_field>
    </settings>
</element>
```

Histogram component

Class name: Histogram

Configuration	Description	Default value*	Valid values
show_grid_lines	Shows the grid Restriction: The grid is only shown if you enabled the display of both axis labels.	false	true false
class_count	Number of displayed classes. If nothing is specified, the HYDRA server calculates the number.		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
show_y_axis_desc	Shows the Y-axis	false	true false
show_x_axis_desc	Shows the X-axis	false	true false
background_color	Background color	R=255,G=255,B=255	
show_chart_caption	Shows the control chart description	true	true false
header_visible	Shows header	false	true false
chart_data_source	Use these fields to request the relevant control chart data from the HYDRA server. If nothing is specified here, the system uses the values from the global option <default_histogram_context>.	RECTYP|PANNR|PAUNR|AFO	

Sample configuration:

```xml
<element class="Histogramm" col="4" colspan="1" row="6" description="Histogram">
    <settings>
        <workdir>Histogramm</workdir>
        <context_change>characteristic</context_change>
        <show_intervention_limit>false</show_intervention_limit>
        <show_x_axis_desc>false</show_x_axis_desc>
        <show_y_axis_desc>true</show_y_axis_desc>
        <show_chart_caption>false</show_chart_caption>
        <background_color>R=213,G=224,B=245</background_color> <!--for example light blue: R=213,G=224,B=245-->
        <class_count>7</class_count>
        <chart_data_source></chart_data_source>
        <header_title></header_title>
        <header_visible>false</header_visible>
        <show_grid_lines>false</show_grid_lines>
        <header_font_style>bold,italic</header_font_style>
    </settings>
</element>
```

Document view showing documents directly

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to *grid* for this presentation component. Generally possible parameters: grid = List of documents immediately showing the selected document list_only = List of documents without showing the selected document immediately file = Directly showing the document with a fixed position number		grid
header_title	Caption		
can_zoom_document	Specifies if the zoom button is shown. If this parameter is not specified, the system checks the global option <can_zoom_document>.	True	true false
header_visible	Shows header	false	true false

header_font_style	Header is in bold and/or italic		Italic bold italic, bold
entry_id	ID of the document to be shown. Refers to the *DOKNR* field in the list of documents.		
list_static_parameter	The static parameters mentioned in "List,56“ in the section "list components" are available.		

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Sample configuration:

```
<element class="DocumentControl" col="5" colspan="2" row="1" rowspan="4" description="document list with displaying a document">
        <settings>
            <workdir>documents</workdir>
            <context_change>characteristic</context_change>
            <document_source>grid</document_source>
            <header_visible>false</header_visible>
            <view_align>vertical</view_align>
            <list_source>LIST;56</list_source>
            <list_dynamic_filter>RECTYP=%RECTYP%|BER=%BER%|PANNR=%PANNR%|PAUNR=%PAUNR%|AFO=%AFO%</list_dynamic_filter>
            <list_static_parameter></list_static_parameter>
            <columndef_section_name>QM-documents_EQD</columndef_section_name>
            <percentage_grid_usage>60</percentage_grid_usage>
            <can_zoom_document></can_zoom_document>
        </settings>
</element>
```

List of documents without showing the document

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to *list_only* for this presentation component. Generally possible parameters: grid = List of documents immediately showing the selected document list_only = List of documents without showing the selected document		list_only

	immediately file = Directly showing the document with a fixed position number		
columndef_section_name	Refers to the column definition from ctaiplay.ini		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
list_grid_local_filter	Filter applied to the grid after requesting and preparing data.		
header_visible	Shows header	false	true false
list_static_parameter	The static parameters mentioned in "List,56“ in the section "list components" are available.		

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Lists with document view

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to *file* for this presentation component. Generally possible parameters: grid = List of documents		file

	immediately showing the selected document list_only = List of documents without showing the selected document immediately file = Directly showing the document with a fixed position number		
columndef_section_name	Refers to the column definition from ctaiplay.ini		
percentage_grid_usage	States the percentage for zooming in/out the document grid	50	
view_align	Aligns the elements (above or below each other)	Vertical	vertical horizontal
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
can_zoom_document	Specifies if the zoom button is shown.		
header_visible	Shows header	false	true false

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

List components

Class name: Grid

Configuration	**Description**	**Default value***	**Valid values**
columndef_section_name	Refers to the column definition from the file		

	"ctaiplay.ini"		
list_static_parameter	Static parameters for the HYDRA server command to be executed		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
header_visible	Shows header	false	true false

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Static elements

Static elements are configured exactly as all other available elements. The only difference is that they are configured in the global configuration file in the element *static_elements.*

In the <*static_elements*> section configure elements that should never be shown with changed content irrespective of the node. One example is the continuous display of a drawing. A drawing can be displayed unchanged irrespective of the current node. The advantage of static elements is that you only have to request these elements once. If you change nodes in the inspection list, the system does not have to request and refresh this element data once more. Below is an example for the definition of the static display of a document with item number 99:

```xml
<static_elements>
    <element class="DocumentControl" col="5" colspan="2" row="5" rowspan="2">
        <settings>
            <workdir>static_document</workdir>
            <document_source>file</document_source>
            <context_change>inspection_requirement</context_change>
            <header_title>Dokument feste Position</header_title>
            <header_visible>false</header_visible>
            <entry_id>99</entry_id>
            <list_source>LIST;56</list_source>
            <list_dynamic_filter>RECTYP=%RECTYP%|BER=%BER%|PANNR=%PANNR%</list_dynamic_filter>
            <columndef_section_name>QM-Merkmalsdokumente_Zusatz</columndef_section_name>
        </settings>
    </element>
</static_elements>
```

OCX extension component

An extension component can include all configurations from the *General Element Configurations*. Further options are managed within the component.

Technical implementation for customer extensions:

The component must support the following functions:

- AIPDataChanged([in] BSTR value);
    - No return value
    - Data are transmitted as a string.
    - Object Pascal (Delphi) procedure AIPDataChanged(const Value: WideString);

  The data string includes the individual fields (values) separated by a pipe. The function is called up when the context changes.
  Configure the context change in the corresponding element using the following XML configuration:
  <context_change>characteristics_value</context_change>

- BeforeRemoveElement
    - No return value
    - No data transmission

  The function is called before the element is destroyed.

Creating an OCX component in Delphi XE7

Start by setting up a new *ActiveX-Library* type project.

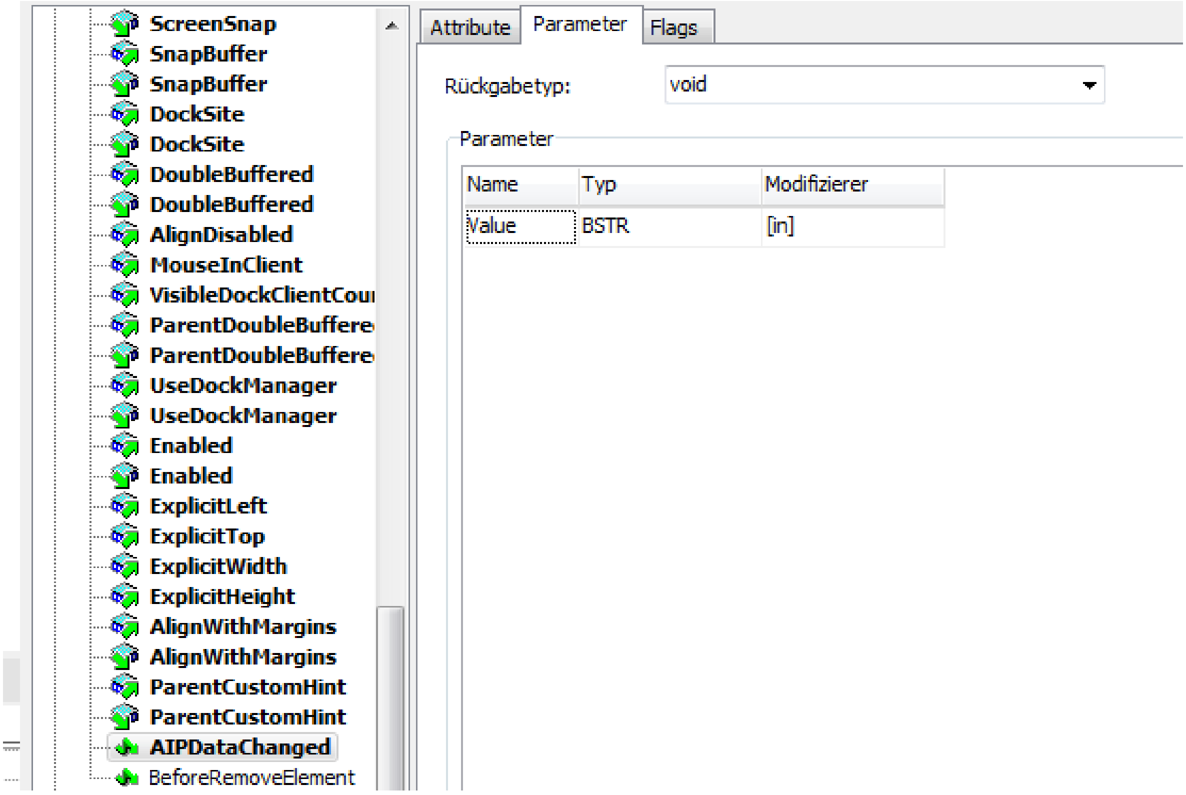
Then, store the project with a relevant name. Do this in the menu *File --> Save project as.* You are prompted to store three files.

In the next step, add an ActiveX form to the project. Go to the menu item *File|New|Other.*

Define the name of the *CoClass*. This name is equivalent to the COM object name in which the interface can be implemented.

The relevant interface is created after clicking *Confirm*. The system implements the functions in the generated form class.

Add the functions listed above. In addition, use the tab button to define the corresponding function parameters (see illustration).



Finally, update the implementation. The system adds the new functions to the form class.

An ActiveX component with the name *Project_name*.ocx is created after clicking on Save. Register this component as administrator with *regsvr32.*

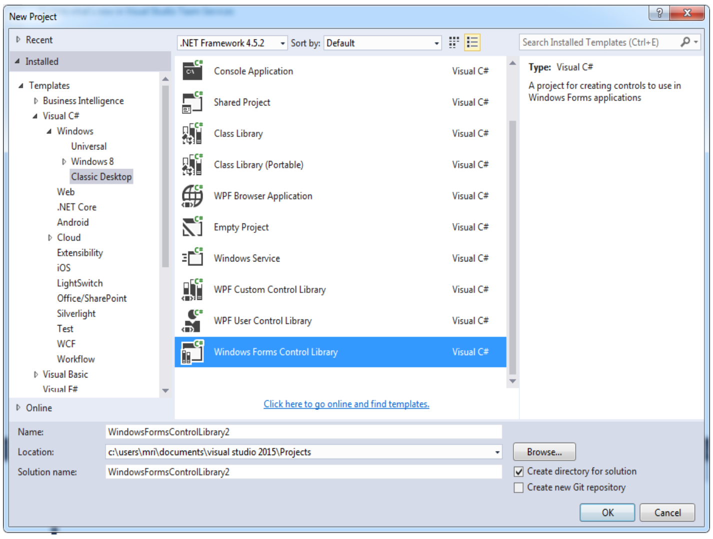
Now integrate this component as an element in the configuration for the relevant measured value dialog. The class name matches the name of the ActiveX Library and the coClass name.

In our example, it is *ExampleOCX.ExampleDisplay*Class.

```
<element class="ExampleOCX.ExampleDisplayClass" col="5" colspan="2" row="1" rowspan="2">
        <settings>
                <context_change>characteristics_value</context_change>
        </settings>
 </element>
```

Creating in Visual Studio 2015

Create a project *Windows Forms Control Library*.



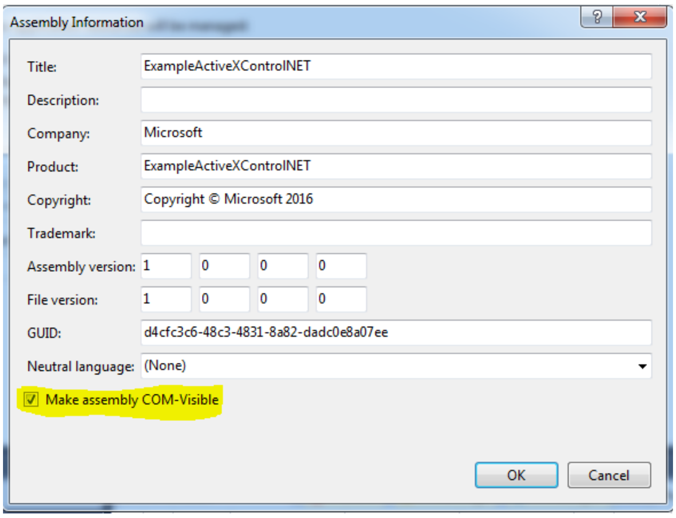
Add the following attributes:

- ComVisible(true)
- Guid → should be re-generated
- ProgId → name with which the COM class is accessed
- ClassInterface(ClassInterfaceType.AutoDual)

In addition, add the functions mentioned in the section "Technical implementation for customizations".

```
namespace ExampleActiveXControlNET
{

        // WICHTIG für COM:
        // ProgId ist für VC++ relevant
        // VB6 orientiert sich am Assemblynamen
        // am besten man benennt Assembly und ProgId gleich
        [ComVisible(true)]
        [Guid("1E173BB5-3C2A-4928-B029-C7F29DC8CEA0")] // GUID muss evtl. neu generieren
        [ProgId("ExampleAxtiveXControlNET")]
        [ClassInterface(ClassInterfaceType.AutoDual)]
        2 references
        public partial class ExampleAxtiveXControl: UserControl
        {
            0 references
            public ExampleAxtiveXControl()
            {
                InitializeComponent();
            }

            0 references
            public void AIPDataChanged(string value)
            {
                label2.Text = value;
            }

            0 references
            public void BeforeRemoveElement()
            {

            }

    }
```

In the next step, make the assembly COM visible in the project properties.



Finally, register the DLL with *regasm.exe*. Use the command:

regasm.exe /codebase mylib.dll

When configuring static elements, make sure that they are only used in one scope! Otherwise there may be side effects when positioning and displaying individual elements as a result of merging the global XML file.

HYDRA lists

Designation	Call parameters
Inspection order header data	List;54 Pruefschritt.lst
Characteristic data	List;55 Merkmal.lst
Further applicable documents/texts	List;56 data.lst (in the folder according to the specification in the relevant XML configuration file) The following "list_static_parameters" are available: • MOD:KONTEXT=PAN Only shows the documents of the inspection requirement. • MOD:KONTEXT=PAUMM Only shows the documents of the characteristic. Filter dynamically by the "OP sequence" and do not use the above-mentioned static parameters in order to show the documents of the characteristic and the inspection requirement.
Failure type catalog	List;57
Measures catalog	List;58
Measured values	List;60 Einzelwerte.lst
Sample status	List;65
Assessment catalogs	List;70
Control chart data	List;71

	RGK.lst
List with results of the inspection order search	List;72
Histogram data	List;79 data.lst (in the folder according to the specification in the relevant XML configuration file)
INI entries from DB	List;95
List of logged on inspection orders	List;96
Statistics calculation	List;99 “AFO”_char_var_stat.lst, e.g. 100_char_var_stat.lst
Analysis selection catalogs	List;100 MerkmalAnAusKat.lst
CAQ options	List;101
Number pool	List;111
Tools	List;112
Cavities	List;122
Samples	List;123 Stichproben.lst
Inspection points	List;124 PPunkt.lst
Failures and measures relating to a machine	List;127

Validation of the configuration

You can use a program to validate the configuration changes that you have made.
The program *configurationvalidator.exe* is stored on the HYDRA server in the subfolder *.\ctnet\win\shared*. Before using the program, make a local copy of the program on your host.

After program start, use the field *Select AIP root directory* to select the AIP directory that contains the changed configurations. Use the AIP main directory and not the subdirectory *.\functions.*
Select in field "*Group"* the entry "*CAQ"* and in field "*Category"* the entry "*EQD".*

Use the button *Validate configuration* to validate all included EQD configurations. The list at the bottom of the window shows the validation results.

Defective entries are identified via a red symbol. Messages have a blue symbol. Validated configuration files that do not include any messages or errors are identified via a green check.

You can ignore error messages with the additional information "-> *The defined class <???_inactive> could not be found*“.

The error message refers to inactive areas that you can easily integrate into you configuration. Just remove the „*_inactive*“ in the class name "*class=*".